

Lecture 2: Verbal and Nonverbal Communication

**COMP3018/COMP5018: Human-Robot
Interaction**

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Semester 2, 2025-2026



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Introduction

- Today's Topics

- 1- Nonverbal communication
- 2- Verbal communication

- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe nonverbal communication and its categories
- 2- Describe verbal communication and speech recognition process
- 3- Discuss the difference between verbal and nonverbal communication

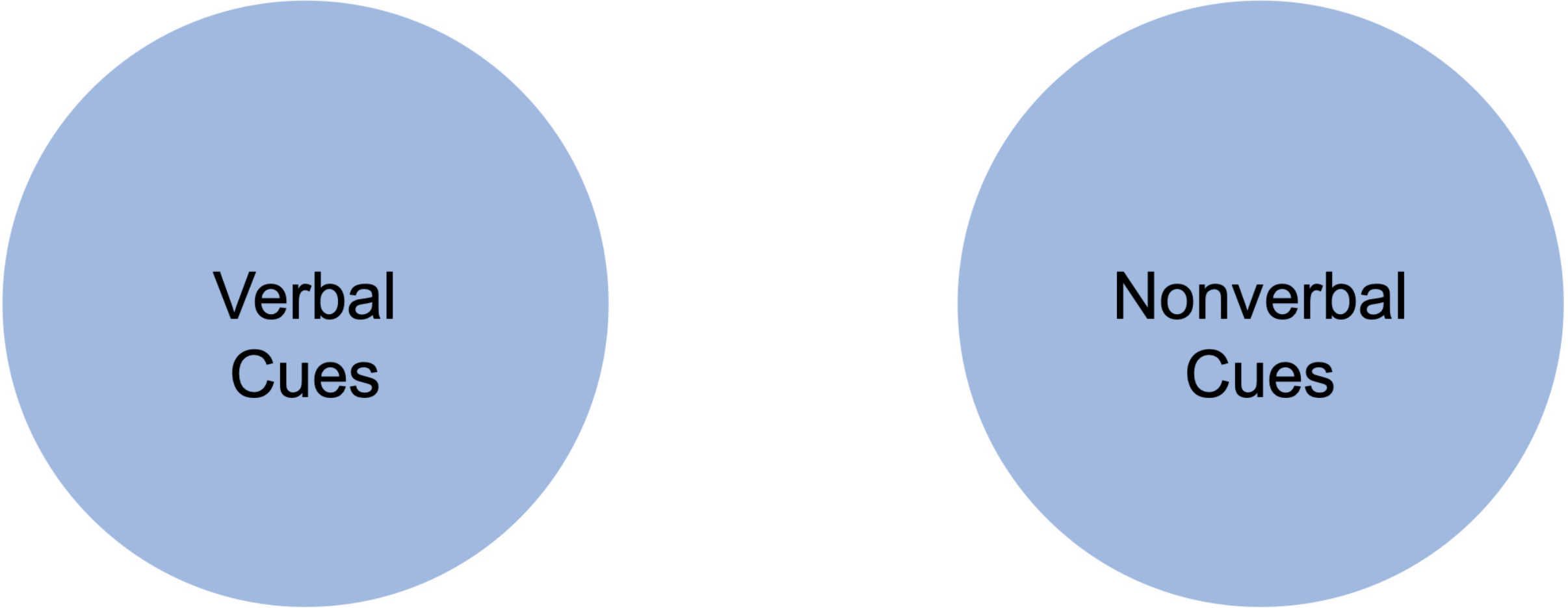
Communication

- **Communication** is simply the act of transferring information from one place, or one person, to another.



- **Communication** can be categorized into **verbal communication** and **non-verbal communication**.

Social Cues



Verbal Cues

- Spoken language (**Speech**)
- Written language

Nonverbal Cues

- Facial expressions
- Gestures
- Postures
- Eye gaze
-

Nonverbal Communication

What is nonverbal communication?

Nonverbal Communication = Communication without words

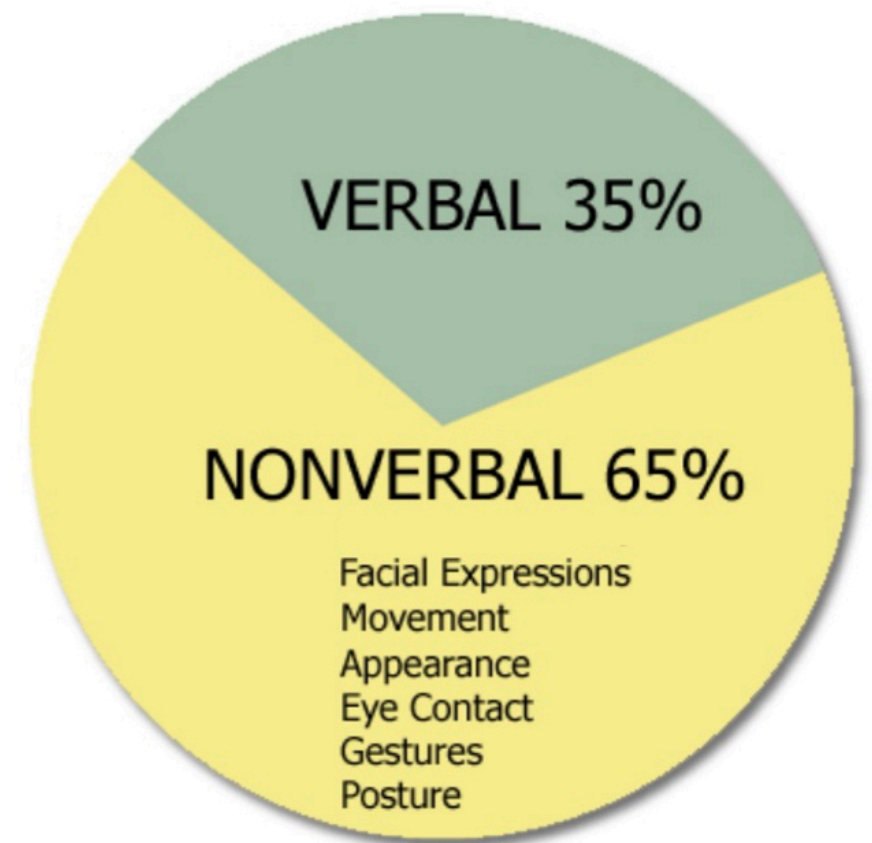
Nonverbal communication is a process of communication through sending and receiving wordless messages.



Nonverbal Communication

- Verbal & nonverbal communication plays an important role in how people interact with one another. People are using around **35% of verbal communication** and **65% of nonverbal communication** in daily life. Nonverbal communication has also cultural meaning.

“The most important thing in communication is to hear what isn't being said.”
Peter F. Drucker



Nonverbal Communication

- Forms of nonverbal communication:

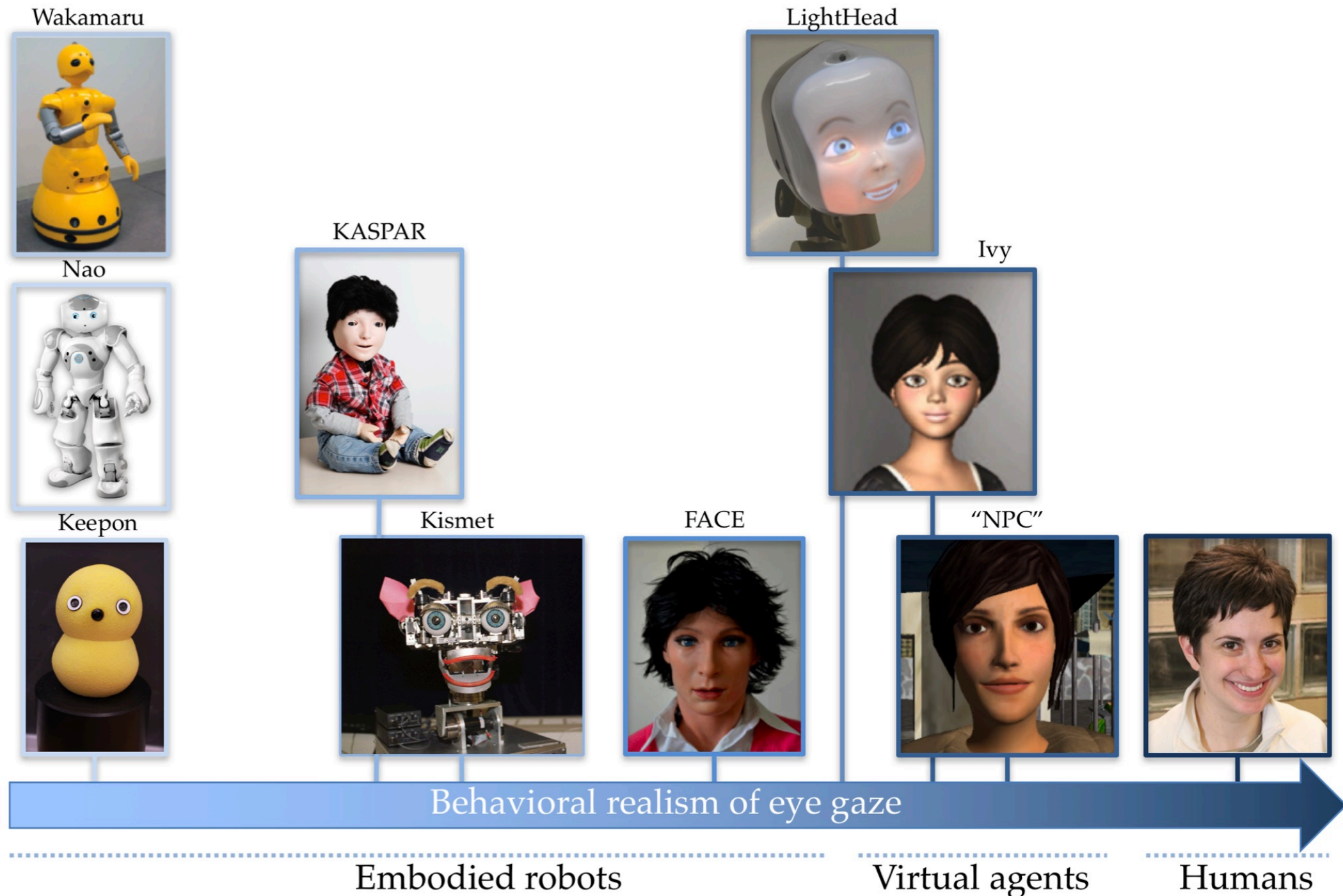
1. Eye Gaze and Movement
2. Gestures
3. Haptics or Touch
4. Posture and Movement
5. Facial Expressions



Nonverbal Communication

- Forms of nonverbal communication:

- Eye gaze and movement



Nonverbal Communication

- **Forms of nonverbal communication:**

1. **Eye gaze and movement**

Gaze is a subtle and important cue for **managing** social interaction

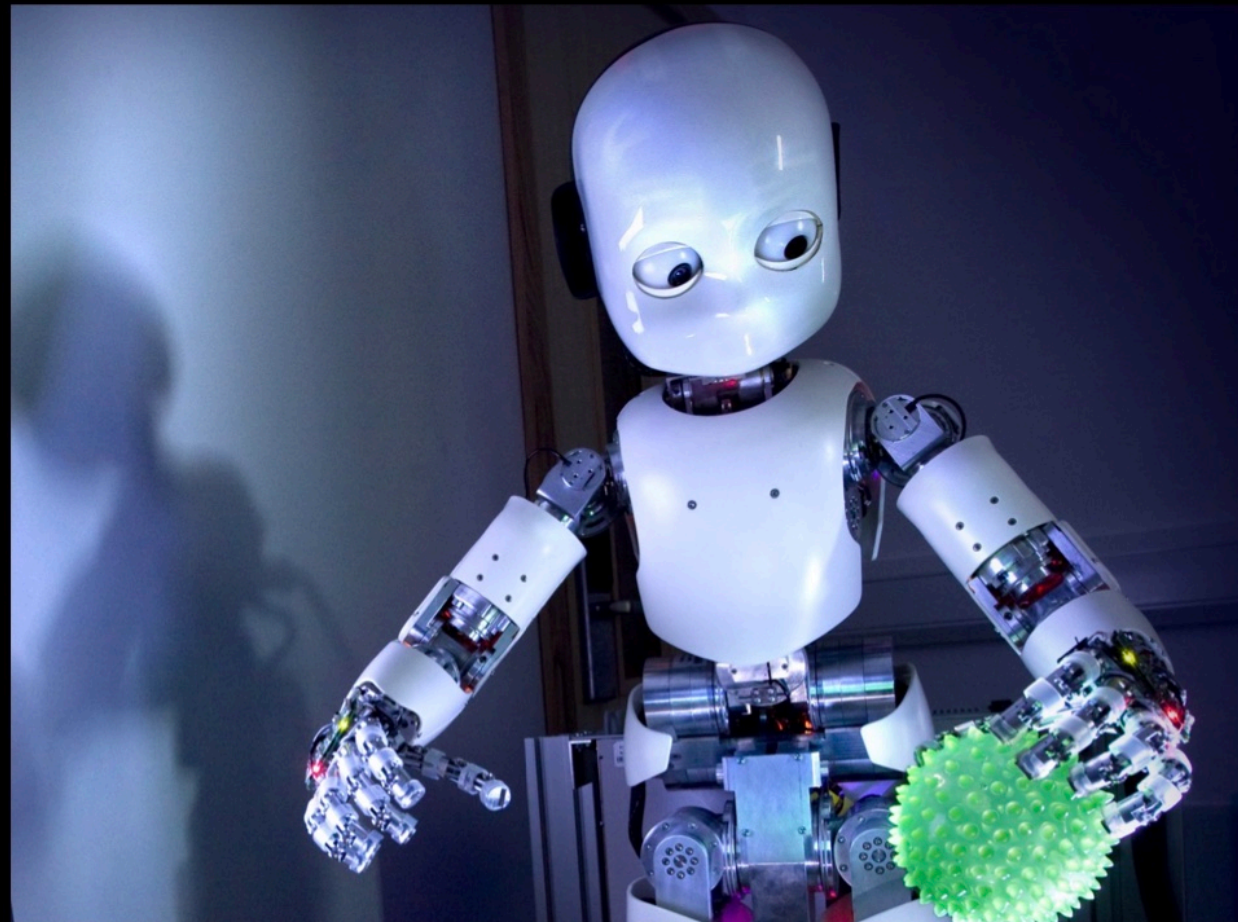
- Gaze **signals** can reflect:

1. Interest
2. Understanding
3. Attention
4. Ability to follow a conversation
5. Willingness to follow a conversation

Nonverbal Communication

- Forms of nonverbal communication:

1. Eye gaze and movement



(Bartneck et al., 2020)

What do you remark about this picture?

Nonverbal Communication

- Forms of nonverbal communication:

1. Eye gaze and movement

Gaze is a subtle and important cue for **managing** social interaction

- Gaze **solicits** and keeps attention
- Gaze **signals** transitions in **turn-taking** (*)
- Gaze is a mechanism for **joint attention** (**)
 - **Timing** and **synchrony** of gaze behavior are important

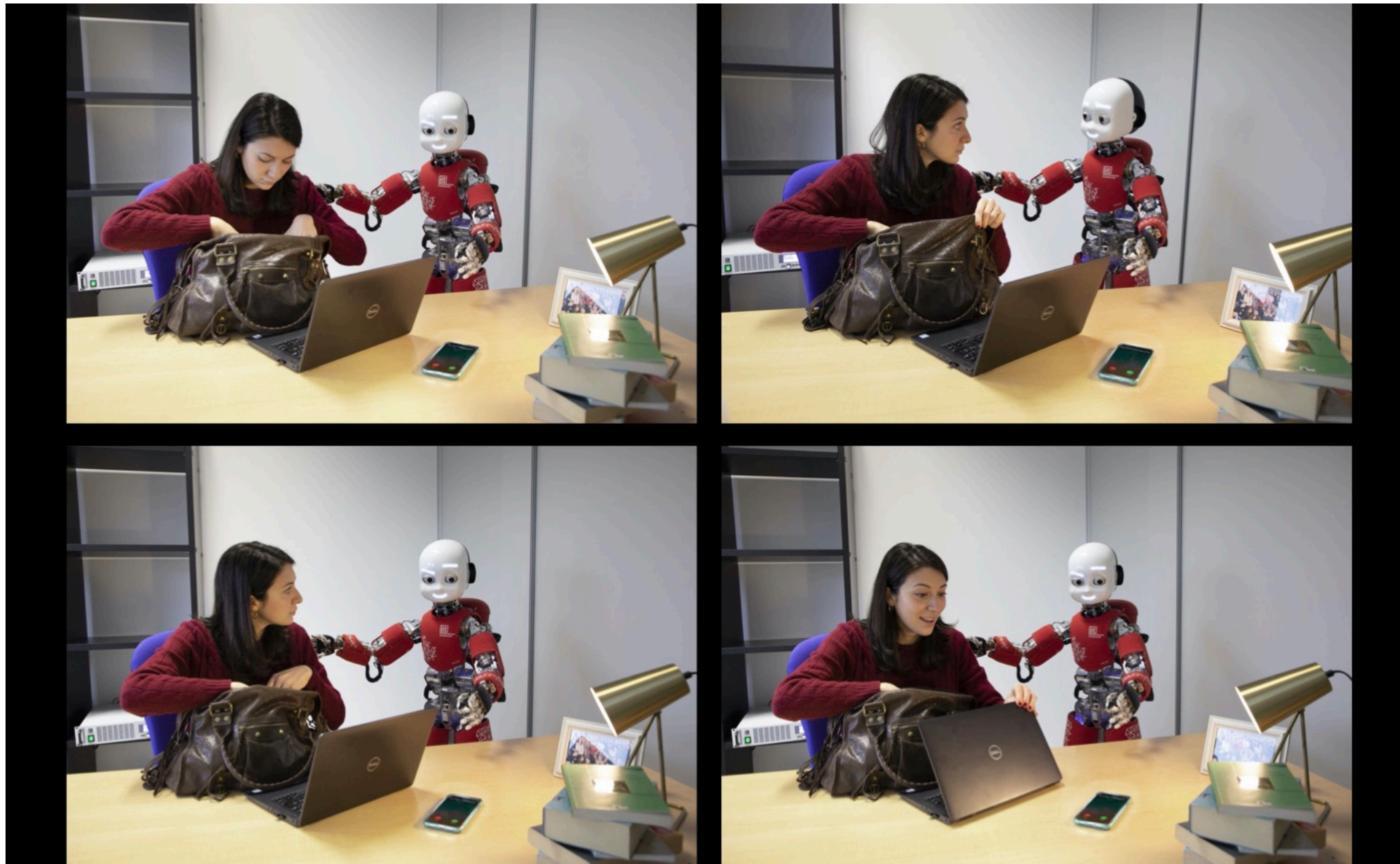
(*) **Turn taking**: refers to the process in which robots **switch between performing actions** and **waiting for other robots or humans to complete their actions**. It is a method used in multi-agent systems to coordinate and synchronize the actions of multiple robots.

(**) **Joint attention**: refers to the ability of multiple robots to **focus their attention on a shared object or task**. It enables robots to coordinate their actions, understand the actions of others, and share information and resources.

Nonverbal Communication

- Forms of nonverbal communication:

1. Eye gaze and movement



"This sequence of pictures depicts a situation in which the iCub humanoid robot (www.icub.org) is interacting with a human, reading her intention to get her phone from her bag, and alerting her to the fact that it is on the desk, hidden from her by the laptop. Note that this sequence has been staged to illustrate the desired capabilities of a cognitive robot and has not yet been implemented."

Sandini, G., A. Sciutti, and D. Vernon, Cognitive Robotics. In M. Ang, O. Khatib, and B. Siciliano (Eds.), Encyclopedia of Robotics. Springer, 2021.
Images courtesy of Istituto Italiano di Tecnologia

Nonverbal Communication

- Forms of nonverbal communication:

2. Gestures

- **Deictic** gestures
 - Pointing to specific things in the environment
- **Iconic** gestures
 - Spreading arms when speaking about something large in size
 - Making a heart shape with two hands
- **Symbolic** gestures
 - Waving hello or goodbye
- **Beat** gestures
 - Accentuate the rhythm of speech

Nonverbal Communication

- Forms of nonverbal communication:

2. Gestures

In HRI, the robot gestures have to be

- Natural
- Well-timed
- Smooth



Nonverbal Communication

- **Forms of nonverbal communication:**

3. Haptics or Touch

Haptic communication is communicating by touch and is used when we come into **physical contact** with others (agents).

- Both deliberate and incidental touch can have **beneficial effects**
 - especially if the interaction partners are from the same social group
- Touch can have **negative effects**
 - especially if the interaction partners are from different social groups
- Touch is an **integral part** of natural human-robot interaction
 - Functional tasks, such as handover and manipulation
 - Social tasks such as **handshaking** (or fist-bump) when greeting

Nonverbal Communication

- Forms of nonverbal communication:

4. Posture and Movement

Posture refers to the position of the body in space, and movement refers to the change in the position of the body over time.

- Posture can be used to **interpret** a person's emotional state

Slow movement

Dropping shoulders

Lethargic gestures



Upright bearing

Fast movements



- Particularly important if the person's face is not visible

Nonverbal Communication

- Forms of nonverbal communication:

4. Posture and Movement

Posture refers to the position of the body in space, and movement refers to the change in the position of the body over time.

- Posture can be used to **signal** a person's disposition:

- Attention
- Engagement
- Attraction

- Defensive posture: holding arms in front of the body



- Positive posture: open arms; hands behind back



Nonverbal Communication

- Forms of nonverbal communication:

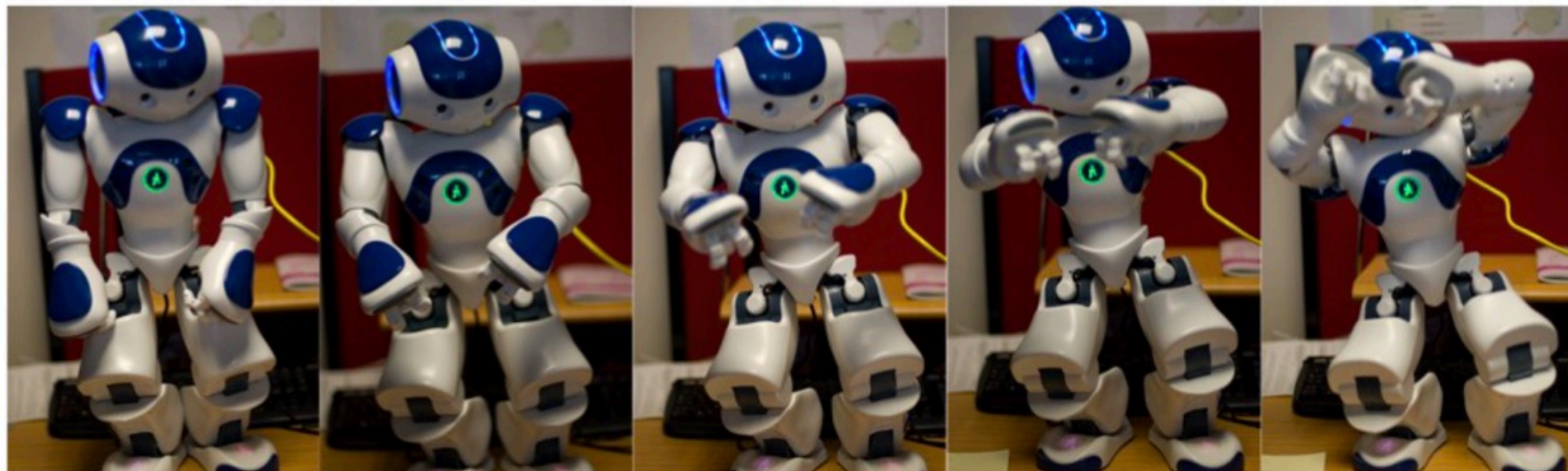
4. Posture and Movement

Posture refers to the position of the body in space, and movement refers to the change in the position of the body over time.

- **Micromovements** can convey the impression that a robot is lifelike

Often implemented as small, random perturbations of the robot's actuators

- The amount and speed of movement can also signal the robot's (simulated) emotional state



Nonverbal Communication

- Forms of nonverbal communication:

5. Facial Expressions

Facial expressions are the key characteristics of nonverbal communication. Facial expressions can communicate different emotions like happiness, sadness, anger, fear, etc.

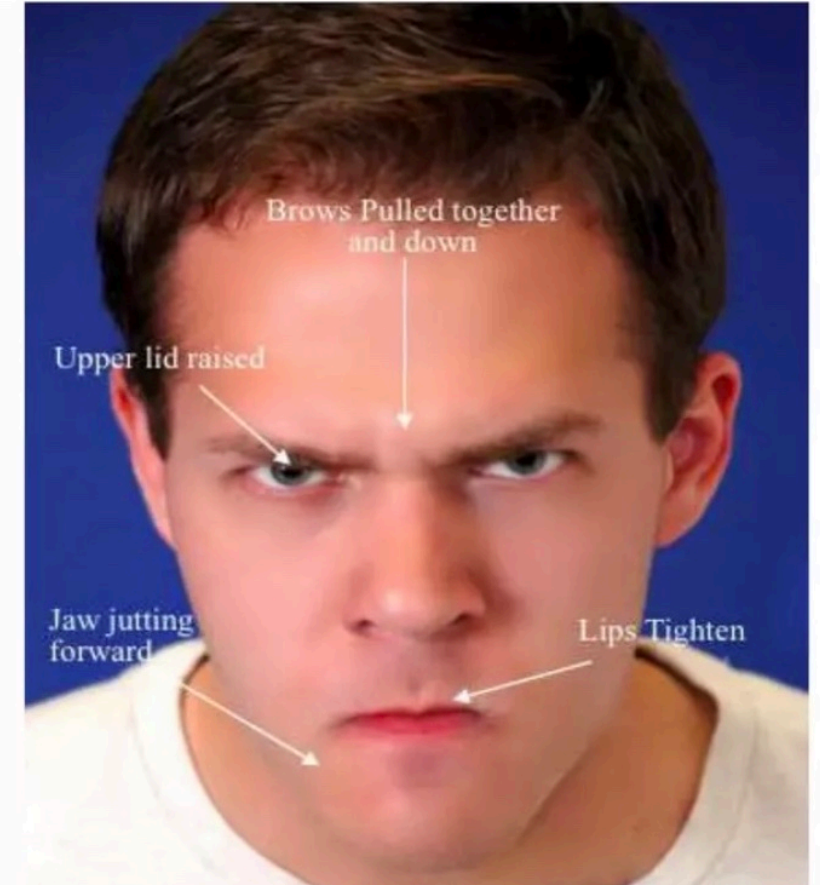


Nonverbal Communication

- Forms of nonverbal communication:

5. Facial Expressions

Facial expressions are the key characteristics of nonverbal communication. Facial expressions can communicate different emotions like happiness, sadness, anger, fear, etc. We can make emotion recognition through **FACS coding**.

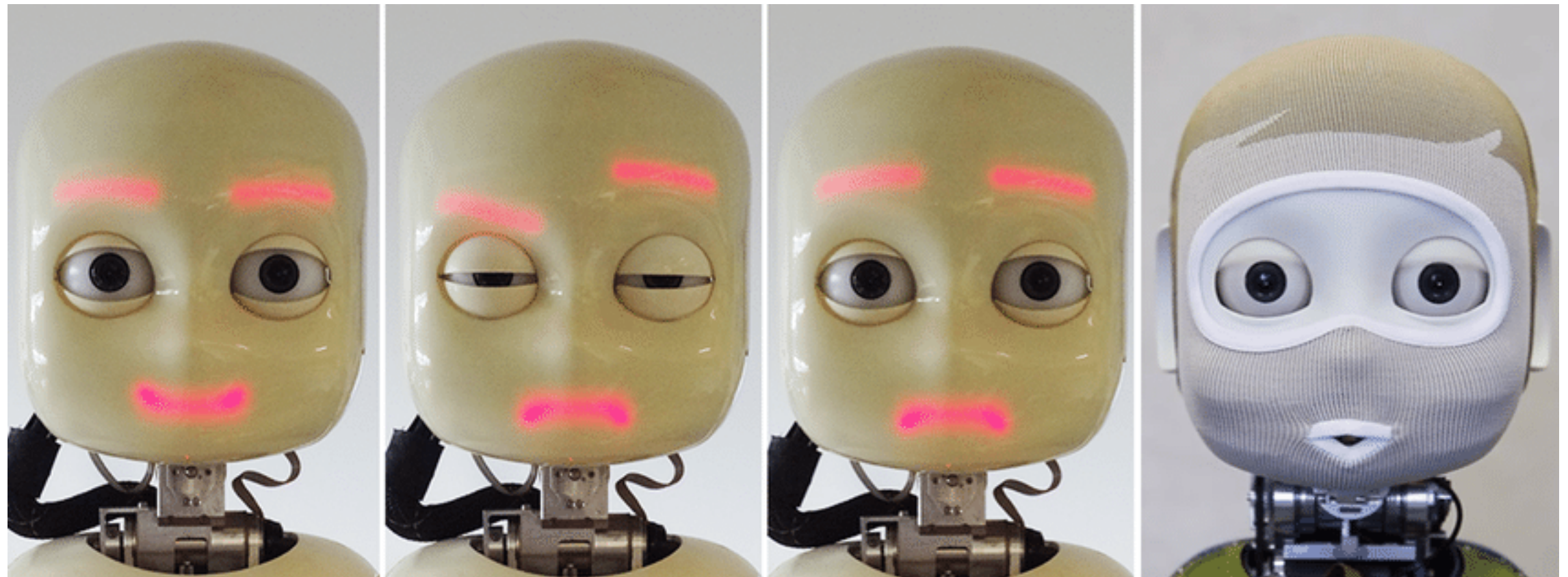


Nonverbal Communication

- Forms of nonverbal communication:

5. Facial Expressions

Facial expressions are the key characteristics of nonverbal communication. Facial expressions can communicate different emotions like happiness, sadness, anger, fear, etc. **Modeling credible facial expressions on robots can be difficult.**



Verbal Communication

◎ Speech is used to

- Convey information
- Create joint attention
- Create a shared reality through communication

◎ Verbal communication is enriched by paralinguistic information

- Prosody (rhythm, timing, ...)
- Non-verbal behavior:
 - Gaze
 - Gestures
 - Facial Expressions
 -

Verbal Communication

- Speech comprises **utterances**: the smallest unit in spoken language
 - **Words**: the smallest units of sound we can utter to convey meaning
 - **Phonemes**: small units of sound that makeup words
 - ❖ Changing [p] to [b] changes the meaning of the word
 - **Pauses**
- Speech also comprises **conversational fillers**
 - They **keep the conversation going**
 - Active listening; they indicate you are paying attention
 - They serve to stop others taking turns and **keep control of the conversation**
 - By leaving no pauses where the other person can interject and take a turn
 - "You know, I was, like, yeah, going to buy her, you know, something, but then I had, like, uhm, what the heck."

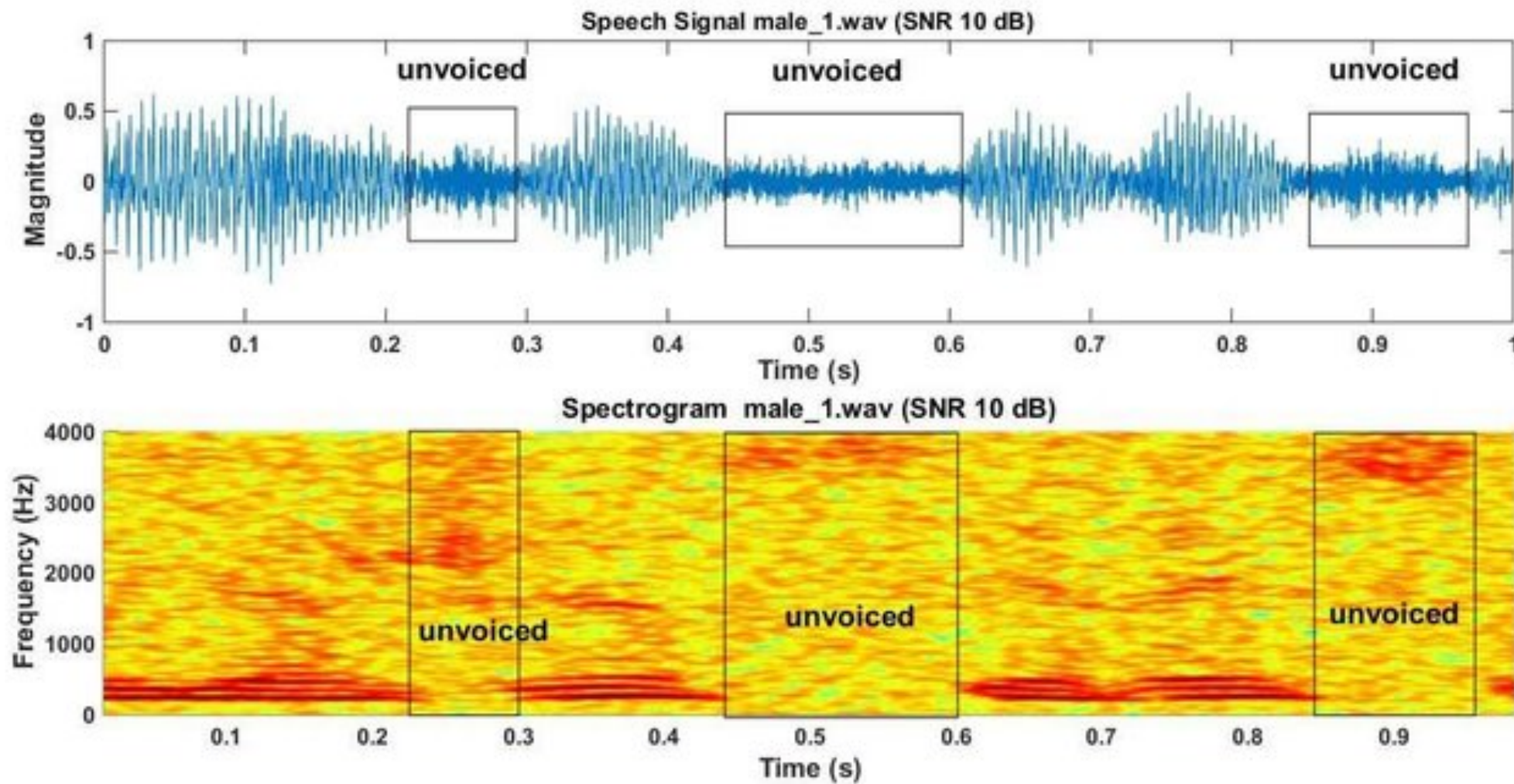
Verbal Communication

- **Written text vs spoken language**
 - Natural, humanlike communication is crucial for HRI
 - Technical prerequisites
 1. Transcribe the speech into words
 2. Demonstrate that it (the machine/robot) understands the words by coming up with appropriate responses
 3. Generate spoken language
- **Speech recognition: (Automated Speech Recognition: ASR)** often makes unrealistic assumptions:
 - Speech is recorded with a high-quality microphone
 - The microphone is positioned close to the speaker
 - In a relatively noise-free environment
- Digital signal processing (**DSP**) and microphone arrays can help

Verbal Communication

- **The speech recognition process:**
 - Digital recording of speech
 - In the time domain
 - Sample the amplitude of the signal over time
 - Convert to the frequency domain
 - Make explicit the strength of certain frequencies
 - Phonemes have different characteristics in the frequency domain, which makes recognition easier.

Verbal Communication



- **Voiced signals:** are produced when the vocal cords vibrate producing sound. Most vowels and some consonants, such as "b," "d," "g," "l," "m," "n," "r," "v," and "w," are produced using voiced speech.
- **Unvoiced signals:** the vocal cords do not vibrate, and there is no resonant quality to the voice. Instead, speech sounds are produced by the movement of air through the mouth and nose. Some consonants, such as "f," "h," "k," "p," "s," "sh," and "t," are produced using unvoiced speech.

Verbal Communication

- Classic approach: extract phonemes, words, sentences

- Using Gaussian Mixture Models (GMM)
- Hidden Markov models

Examples of models but deep models can be also used

- Use them to generate a **probabilistic model** how phonemes are strung together to form valid sentence
- "**robot**" is more likely than "**lobot**"
- "**the robot served the man**" is more likely than "**the robot swerved the man**"

- **Limitations of Speech Recognition**: All speech-recognition engines still struggle with **atypical speech** that might occur due to insufficient training on:

- Young speakers
- Elderly speakers
- Local dialects
- Nonnative speakers

Summary

- Today we discussed

- 1- Nonverbal communication

- 2- Verbal communication

- Next week

- 1- We will discuss a different topic in social robotics; i.e., Emotional Intelligence.

Readings

- C. Bartneck and T. Belpaeme, et al., "Human-Robot Interaction", Cambridge University Press, 2020.